Hadi Reisizadeh

Postdoctoral Associate, Optimization for AI Lab, University of Minnesota

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Summary

Research Scientist specializing in **LLM safety, optimization, and trustworthy AI**. Exposed a critical vulnerability in LLM unlearning: sensitive knowledge reliably resurfaces under probabilistic decoding. Introduced **leak@**k, a scalable meta-metric to quantify worst-case knowledge leakage, and **RULE**, an iterative training framework that achieves near-zero leakage on standard benchmarks. Developed **BLUR**, a bi-level optimization approach delivering up to **19**% stronger unlearning while preserving model utility on LLaMA-2-7B and Zephyr-7B-Beta. Focused on building AI systems whose behavior can be verified, corrected, and trusted.

Technical Skills

Core Expertise	LLMs, optimization, machine unlearning, efficient training, and safe AI
Programming Languages	Python, MATLAB, R, C/C++
Machine Learning	PyTorch, HuggingFace Transformers, TensorFlow, Scikit-learn
LLM Training	DeepSpeed, HuggingFace Accelerate

Work Experience

University of Minnesota, Minneapolis, MN

Postdoctoral Associate, Optimization for AI Lab — 09/2024 – Present

Research Assistant, Information Processing Group — 09/2017 – 09/2024

- **LLM Unlearning and Model Safety.**
  - Explored the fundamental limits of unlearning in LLMs, showing that sensitive knowledge can resurface under probabilistic decoding. Introduced **leak@**k, a meta-metric to quantify worst-case knowledge leakage across k generations, and **RULE**, an iterative training framework achieving near-zero leakage on TOFU and outperforming SOTA methods on MUSE-News [paper], [code].
  - Proposed a **bi-level optimization framework (BLUR)** for LLM unlearning that prioritizes the forget objective over the retain loss, achieving up to **19**% higher unlearning efficiency and **11**.**7**% higher model utility on MUSE-News with the **LLaMA-2-7B** model, and **16**.**6**% higher efficiency on WMDP with equivalent utility using the **Zephyr-7B-Beta** model [paper], [code].
- **Efficient Reasoning Condensation.** Proposed an **edge-preserving condensation framework** for chain-of-thought reasoning training that retains only the initial and final reasoning stages, enabling supervised models to achieve **lossless reasoning accuracy** while reducing training time by over **34**% across multiple **LLM families (Qwen, LLaMA)** and reasoning benchmarks [paper], [code].
- **Distributed Training.** Designed a **two-time-scale optimization algorithm** for communication-efficient distributed training over noisy, time-varying networks, achieving **provable convergence guarantees** in both convex and non-convex settings [paper 1], [paper 2], [paper 3].

Seagate Technology, Shakopee, MN — 05/2023 – 04/2024

Research Intern, Data Trust Research Team

- **Differentially Private Distributed Training.** Developed an **accelerated privacy-preserving distributed training method**, achieving faster convergence without reducing privacy guarantees.

Education

Ph.D. & M.Sc. in Electrical and Computer Engineering (*Minor in Computer Science*)

University of Minnesota, Twin Cities, Minneapolis, MN — 09/2017 – 09/2024

B.Sc. in Electrical and Computer Engineering

Isfahan University of Technology, Isfahan, Iran — 09/2012 – 05/2017

Selected Publications

- **H. Reisizadeh**, J. Ruan, Y. Chen, S. Pal, S. Liu, and M. Hong, *"Leak@k: Unlearning Does Not Make LLMs Forget Under Probabilistic Decoding,"* ICML, 2026.
- **H. Reisizadeh**, J. Jia, Z. Bu, B. Vinzamuri, A. Ramakrishna, K. Chang, V. Cevher, S. Liu, and M. Hong, *"BLUR: A Bi-Level Optimization Approach for LLM Unlearning,"* EACL, 2026.
- J. Jia, **H. Reisizadeh**, C. Fan, N. Baracaldo, M. Hong, and S. Liu, *"EPiC: Towards Lossless Speedup for Reasoning Training through Edge-Preserving CoT Condensation,"* Submitted, 2026.
- **H. Reisizadeh**, B. Touri, and S. Mohajer, *"Almost Sure Convergence of Distributed Optimization with Imperfect Information Sharing,"* Automatica, 2025.
- **H. Reisizadeh**, B. Touri, and S. Mohajer, *"Distributed Optimization Over Time-Varying Graphs With Imperfect Sharing of Information,"* IEEE Transactions on Automatic Control, 2022.